TripEasy Tourism Management System

Day visit planner for Gampaha

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Abbreviations

- **API:** Application Programming Interface
- **CRUD:** Create, Read, Update, Delete
- **CSS:** Cascading Style Sheets
- **REST:** Representational State Transfer
- **SRS:** Software Requirements Specification
- **UI:** User Interface

1.0 Introduction / Background

Ragama, situated in the Gampaha District of Sri Lanka, serves as an accessible hub surrounded by rich cultural, historical, and natural landmarks within a 25 km radius **[1]**, **[3]**. While the region boasts major attractions—ranging from sacred religious sites like the Tewatta Basilica and Kelaniya Temple to serene natural reserves and coastal spots like Negombo Beach **[4]**—tourists often overlook lesser-known cultural and heritage sites. Currently, regional travel information is fragmented across disconnected web pages or completely undocumented online **[1]**. This web application solves this gap by consolidating reliable local travel details, operational hours, distances, and an interactive itinerary planner onto a single platform **[1]**, **[5]**.

1.1 Problem Statement

Tourists visiting the Gampaha region struggle to find organized, accurate travel information regarding local attractions, resulting in inefficient day trips **[1]**, **[4]**. Essential travel details such as opening hours, travel distances, and location-based routing are scattered across unreliable web sources or absent entirely **[3]**. Solving this problem is crucial to promote local tourism, optimize single-day travel routes, and save visitors valuable time **[1]**.

1.2 Aim

To design and develop a web-based local tourism information system that enables tourists to explore attractions within a 25 km radius of Ragama and customize an efficient one-day visit itinerary.

1.3 Project Objectives

- To gather and validate tourist requirements through stakeholder consultations with four team members **[5]**.

- To build an interactive web interface allowing users to view, filter, and search 10 local places of interest by category.

- To integrate an interactive map system displaying place locations, travel distances, and route details within a 25 km radius **[3]**.

- To implement a custom one-day visit planner feature where users can select multiple sites to create a tailored itinerary.

- To develop a secure administrative portal providing full CRUD capabilities over place information.

- To manage code versioning using Git and deploy a fully functional web application **[5]**.

1.4 Scope and Limitations

1.4.1 Scope

- **Location Radius:** Covers 10 selected places of interest within 25 km of Ragama (Gampaha District).

- **Categorization:** Categorizes places under Religious, Nature, Hiking, Heritage, and Cultural themes.

- **Core Features:** Interactive place list for 1 day visit planning, category filtering, detailed views (opening times, travel tips, distance), interactive map markers, user itinerary planning, and admin CRUD portal.

1.4.2 Limitations

- No online booking facilities for tickets or venue access.

- No payment gateway processing or financial transaction management.

- No hotel accommodation or third-party vehicle transport reservation services.

1.5 Target Users

- **Tourist User:** Local and international visitors seeking curated travel information, map navigation, category filtering, and customized one-day trip planning tools.

- **Administrator:** System manager responsible for logging into the backend portal to add, update, or remove place details, photos, operational hours, and category classifications.

1.6 Stakeholder Identification Plan

Stakeholder	Role	Key Requirement Identified
Abdulla MT — E2410182	System Administrator	Dynamic Administrator module with CRUD permissions (add, edit, delete) to manage attraction data and keep details up to date.
Ahamed MHR — E2410181	Tourist	Comprehensive spot profile data including categories, visual media (photos), precise map coordinates, travel tips, and estimated travel times.
Kirushika N — E2410167	Local Visitor	Practical day-visit planning tools including location filtering, travel distances, entrance fee information, and nearby facility details.
Yasvini S — E2410180	Investor	Simple 1-day itinerary creation features including estimated visit durations, transportation advice, and suggested travel routes to optimize regional tourism.

Table 1:stakeholders requirements analysis

1.7 Proposed System Overview

The proposed system operates as a single-page or multi-page web application architecture. Users land on a main discovery dashboard displaying a map alongside place cards categorized by interest. The frontend communicates via RESTful API calls to the backend web server, which fetches data from the database. When a user selects attractions, the frontend dynamically calculates total distances and renders a daily itinerary view. Administrators authenticate through a secure route to manage place records directly in the database.

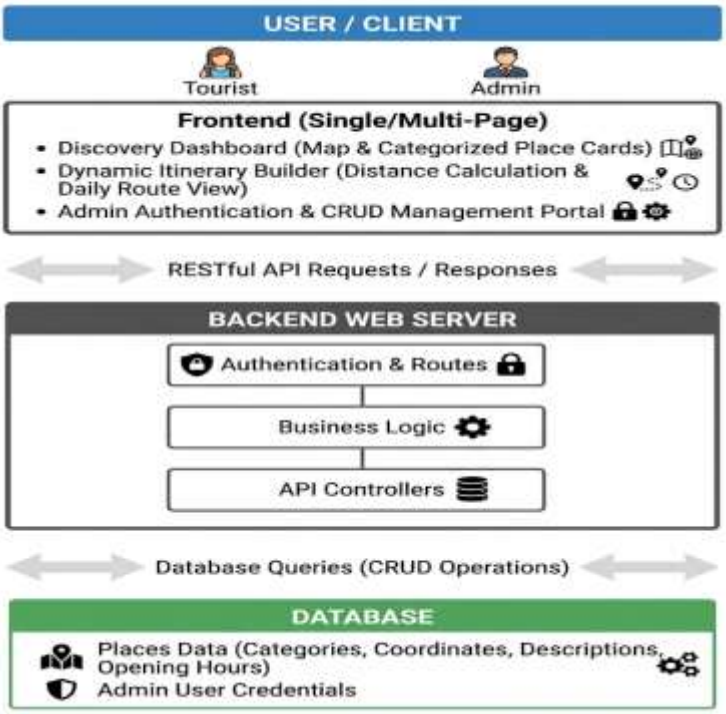


Figure 1: block diagram

1.8 Preliminary List of Places

#	Place Name	Category	Distance from your location	Description of Places
1	National Basilica of Our Lady of Lanka, Tewatta	Religious	~2.5 km	Prominent Catholic pilgrimage shrine featuring a vast indoor sanctuary, stained glass, and tranquil gardens. (Open: 5:00 AM – 6:30 PM)
2	Kelaniya Raja Maha Viharaya	Religious	~10 km	Historic Buddhist temple situated along the Kelani River, celebrated for its stupa and Solias Mendis wall murals. (Open: 5:30 AM – 8:00 PM)
3	Uswetakeiyawa Beach	Nature	~11 km	Serene coastal beach area near Kandana/Wattala, perfect for ocean breezes, sunset walks, and quiet relaxation.

#	Place Name	Category	Distance from your location	Description of Places
4	Gampaha Botanical Garden (Henarathgoda)	Nature	~18 km	Historic botanical garden featuring Asia's first planted rubber tree, shaded paths, and boat rides. (Open: 7:30 AM – 6:00 PM)
5	Maligatenna Raja Maha Viharaya	Hiking	~23 km	Ancient temple on Gampaha's highest peak featuring rock stairs, granite outcrops, and 360-degree summit views. (Open: 6:00 AM – 8:00 PM)
6	Pilikuttuwa Raja Maha Viharaya	Hiking	~18 km	Forest cave complex offering shaded nature through dense canopy and over 90 natural rock caves.
7	Vaarana Rajamaha Viharaya	Heritage	~21 km	Ancient rock cave temple dating back to King Devanampiyatissa, split across three hill levels with old frescoes. (Open: 6:00 AM – 8:00 PM)
8	Kiththa Restaurant	Restaurants	~1.8 km	High-demand family & group dining 100+ Sri Lankan, Chinese, Thai, and Western dishes, AC hall, outdoor garden, and live music. (open 11:00 AM – 11:00 PM).
9	Dutch Canal (Hamilton Canal)	Sightseeing	~8 km	Historic waterway constructed during the colonial Dutch era, stretching near Hendala for scenic local boat rides.
10	Guruge Nature Park	Sightseeing	~7 km	Theme park showcasing traditional Sri Lankan cultural heritage, an ancient village replica, museum, and family rides. (8:30 AM – 6:00PM)

Table 2: Attraction places in Gampaha

1.9 Technology Stack

Software Requirements Specification (SRS) documents and system architectural mockups follow standardized IEEE computer software engineering practices **[5]**.

Frontend: React.js, Tailwind CSS, React-Leaflet

Backend: Node.js, Express.js framework

Database: MongoDB Atlas

Development Tools: Visual Studio Code, Postman (API Testing), Figma (UI Mockups)

Version Control (Git): Git, GitHub

Technology / Tool	Why Choose This
React.js	Delivers a fast, interactive single-page user interface through a reusable, component-based architecture.
Tailwind CSS	Enables rapid, responsive UI design using utility-first CSS classes directly inside components.
React-Leaflet	Provides a lightweight, free, and open-source interactive mapping solution without requiring paid API keys.
Backend: Node.js	Executes JavaScript on the server side to handle asynchronous events and enable scalable full-stack development.
Backend: Express.js	Simplifies backend development by providing a lightweight routing system for handling RESTful API requests.
Database: MongoDB Atlas	Offers a flexible, cloud-hosted NoSQL database that easily stores JSON-like dynamic place data and user records.
Visual Studio Code	Serves as a lightweight, feature-rich code editor with integrated terminal and debugging tools.
Postman	Streamlines backend API development by allowing quick testing and validation of REST endpoints before frontend integration.
Figma	Facilitates modern user interface design and rapid wireframing before starting actual code implementation.
Version Control: Git & GitHub	Manages source code version tracking and provides cloud-backed repository hosting for safe collaboration.

Table 3: Technologies & tools

Conclusion

This project provides a centralized, user-friendly digital platform for discovering hidden and prominent travel destinations within the Gampaha region. By combining interactive mapping, category filtering, and custom itinerary planning, the application significantly enhances the day-visit planning experience for local tourists. Through its completion, practical expertise in requirement elicitation, full-stack software development, database design, and technical presentation will be gained.

References

[1] Sri Lanka Tourism Development Authority (SLTDA), *Annual Performance Report 2024*, Colombo, Sri Lanka: Ministry of Tourism and Lands, 2024.

[2] Department of National Botanic Gardens, "Henarathgoda Botanic Gardens, Gampaha," *Ministry of Botanical Gardens and Public Recreation*, Peradeniya, Sri Lanka, 2024. [Online]. Available: https://www.botanicgardens.gov.lk.

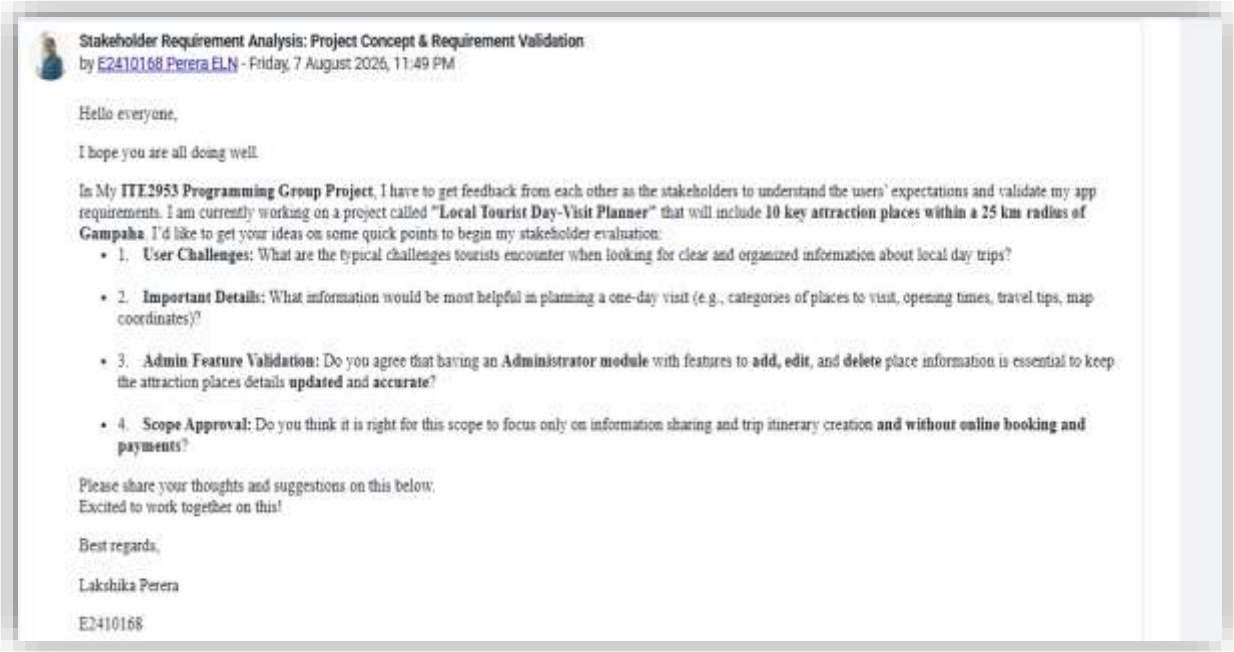
[3] District Secretariat Gampaha, *Gampaha District Tourism Resource Directory*, Gampaha, Sri Lanka: Public Administration & Home Affairs, 2023.

[4] Lakpura LLC, "Gampaha District Travel and Cultural Attractions Guide," *Lakpura Heritage & Tourism*, 2024. [Online]. Available: https://www.lakpura.com/pages/gampaha-district.

[5] IEEE Computer Society, *IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications*, IEEE Computer Society, New York, NY, USA, 1998.

Appendices

Appendix A



Stakeholder Requirement Analysis: Project Concept & Requirement Validation
by E2410168 Perera ELN - Friday, 7 August 2026, 11:49 PM

Hello everyone,

I hope you are all doing well.

In My **ITE2953 Programming Group Project**, I have to get feedback from each other as the stakeholders to understand the users' expectations and validate my app requirements. I am currently working on a project called **"Local Tourist Day-Visit Planner"** that will include **10 key attraction places within a 25 km radius of Gampaha**. I'd like to get your ideas on some quick points to begin my stakeholder evaluation:

- 1. **User Challenges:** What are the typical challenges tourists encounter when looking for clear and organized information about local day trips?
- 2. **Important Details:** What information would be most helpful in planning a one-day visit (e.g., categories of places to visit, opening times, travel tips, map coordinates)?
- 3. **Admin Feature Validation:** Do you agree that having an **Administrator module** with features to **add, edit**, and **delete** place information is essential to keep the attraction places details **updated** and **accurate**?
- 4. **Scope Approval:** Do you think it is right for this scope to focus only on information sharing and trip itinerary creation **and without online booking and payments**?

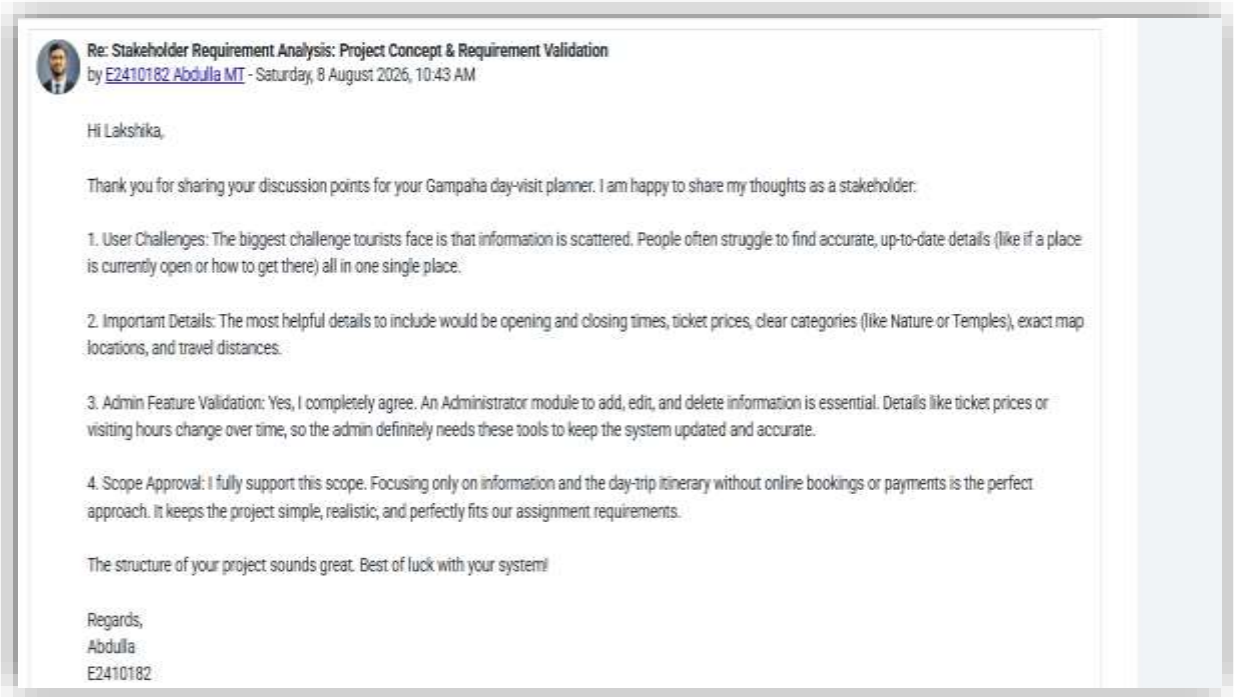
Please share your thoughts and suggestions on this below.
Excited to work together on this!

Best regards,

Lakshika Perera

E2410168

Figure 2: discussion



Re: Stakeholder Requirement Analysis: Project Concept & Requirement Validation
by E2410182 Abdulla MT - Saturday, 8 August 2026, 10:43 AM

Hi Lakshika,

Thank you for sharing your discussion points for your Gampaha day-visit planner. I am happy to share my thoughts as a stakeholder:

1. User Challenges: The biggest challenge tourists face is that information is scattered. People often struggle to find accurate, up-to-date details (like if a place is currently open or how to get there) all in one single place.

2. Important Details: The most helpful details to include would be opening and closing times, ticket prices, clear categories (like Nature or Temples), exact map locations, and travel distances.

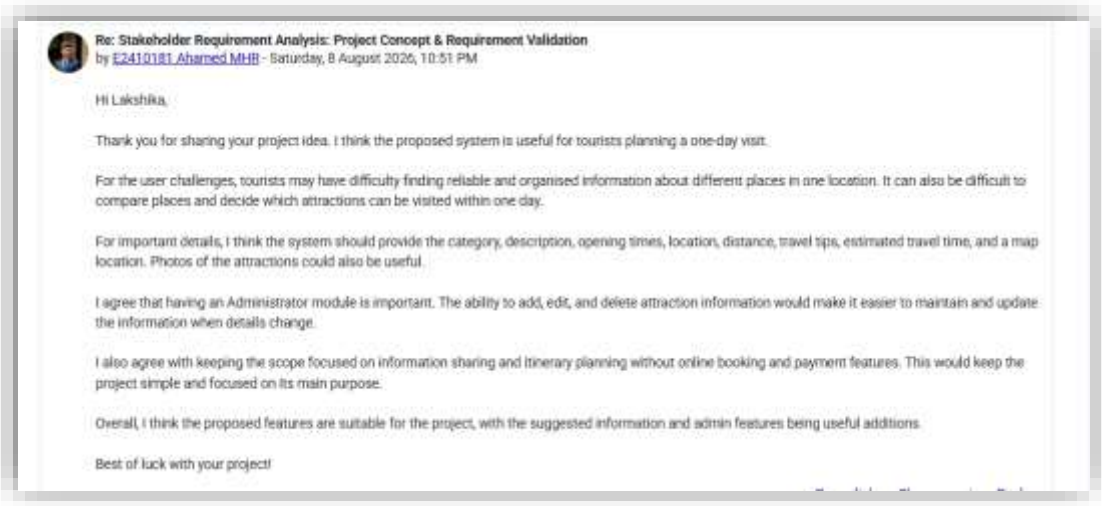
3. Admin Feature Validation: Yes, I completely agree. An Administrator module to add, edit, and delete information is essential. Details like ticket prices or visiting hours change over time, so the admin definitely needs these tools to keep the system updated and accurate.

4. Scope Approval: I fully support this scope. Focusing only on information and the day-trip itinerary without online bookings or payments is the perfect approach. It keeps the project simple, realistic, and perfectly fits our assignment requirements.

The structure of your project sounds great. Best of luck with your system!

Regards,
Abdulla
E2410182

Figure 3: Feedback 1



Re: Stakeholder Requirement Analysis: Project Concept & Requirement Validation
by E2410181 Ahamed MHR - Saturday, 8 August 2026, 10:51 PM

Hi Lakshika,

Thank you for sharing your project idea. I think the proposed system is useful for tourists planning a one-day visit.

For the user challenges, tourists may have difficulty finding reliable and organised information about different places in one location. It can also be difficult to compare places and decide which attractions can be visited within one day.

For important details, I think the system should provide the category, description, opening times, location, distance, travel tips, estimated travel time, and a map location. Photos of the attractions could also be useful.

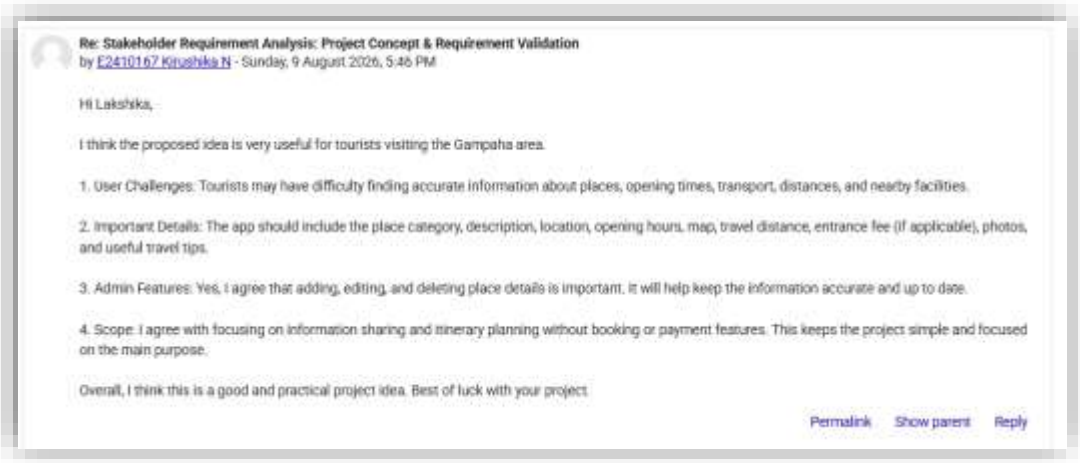
I agree that having an Administrator module is important. The ability to add, edit, and delete attraction information would make it easier to maintain and update the information when details change.

I also agree with keeping the scope focused on information sharing and itinerary planning without online booking and payment features. This would keep the project simple and focused on its main purpose.

Overall, I think the proposed features are suitable for the project, with the suggested information and admin features being useful additions.

Best of luck with your project!

Figure 4: Feedback 2



Re: Stakeholder Requirement Analysis: Project Concept & Requirement Validation
by E2410167 Kirushika N - Sunday, 9 August 2026, 5:46 PM

Hi Lakshika,

I think the proposed idea is very useful for tourists visiting the Gampaha area.

1. User Challenges: Tourists may have difficulty finding accurate information about places, opening times, transport, distances, and nearby facilities.

2. Important Details: The app should include the place category, description, location, opening hours, map, travel distance, entrance fee (if applicable), photos, and useful travel tips.

3. Admin Features: Yes, I agree that adding, editing, and deleting place details is important. It will help keep the information accurate and up to date.

4. Scope: I agree with focusing on information sharing and itinerary planning without booking or payment features. This keeps the project simple and focused on the main purpose.

Overall, I think this is a good and practical project idea. Best of luck with your project.

Permalink Show parent Reply

Figure 5: Feedback 3



Re: Stakeholder Requirement Analysis: Project Concept & Requirement Validation
by E2410180 Yasvini S - Monday, 10 August 2026, 7:02 PM

Hello,

Thank you for sharing your project proposal. I believe this is a useful concept for tourists who are planning a short visit around Gampaha.

From my perspective, the biggest challenge is comparing multiple attractions and selecting the best places to visit within a limited time. The app should provide estimated visit duration, nearby attractions, transportation options, and suggested travel routes in addition to basic place information.

I support the inclusion of an Administrator module, as regular updates will improve the reliability of the information provided. I also think the current scope is appropriate because focusing on itinerary planning and information management makes the project more achievable and easier for users to navigate.

I wish you all the best with your project.

Figure 6: Feedback 4

Appendix B

Project Plan & Gantt Chart

Activity	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
System Requirements Analysis												
Tourist Place Data Collection												
System Design												
Database Design												
User Interface Design												
Frontend Development												
Backend Development												
Administrator Module Development												
Tourist Module Development												
One-Day Visit Planner Development												
Map Integration												
System Testing												
Bug Fixing and Improvements												
Final System Preparation												

Figure 7: project plan with time